

Biological Projects

Please clearly label your file – include the text of the question in your file. Unlike the homework, projects should be written in grammatical sentences and contain an introduction, results, and conclusion/discussion. Make sure to include enough results to justify your conclusions, but do not include figures that you don't discuss. The discussion should include a general description of the salient points of the result, as well as answering any specific questions.

BP1: Dotplots and Alignments

20 pts per target, you may do multiple targets

First find some distantly related target sequences

Perform a blast protein search using one of the following target sequences, or a sequence of your choice. In each case, choose two related sequences so that all three sequences (the two new sequences and your original query) meet the following criteria for all pairs.

- < 35% identity
- Alignment length > 100
- All sequences approximately the same length

Target sequences. Here are some possibly interesting sequences with some suggestions on how to better find distant but matching sequences in parentheses. Feel free to choose your own sequences; contact me if you are unsure whether the sequences are appropriate. Another good strategy for finding distantly related homologs is to do Blast search limited to specific kingdoms of life.

- wheat starch synthase, GI:9502143 (exclude viridiplantae)
- human alpha hemoglobin, GI:4504347 (limit search to bacteria)
- arabidopsis BRI1, GI:332661669 (exclude viridiplantae)
- Drosophila cytochrome P450 reductase GI:17137192 (exclude animalia)
- Sirtuins
- acetyl-CoA decarboxylase/synthase (any subunit)
- MarR transcriptional regulator
- The following paper gives a long list of ancient proteins that would be interesting targets:
Sousa FL, Nelson-Sathi S, Martin WF. One step beyond a ribosome: The ancient anaerobic core. *Biochim Biophys Acta*. 2016 Aug;1857(8):1027-1038. doi: 10.1016/j.bbabi.2016.04.284. Epub 2016 May 2. PMID: 27150504; PMCID: PMC4906156.

Use the following methods to compare the sequences and analyze their similarity (up to 20 pts for each 3-tuple of sequences)

1. Dotplots

- Perform multiple dotplots at different window lengths and cutoffs and identify the best settings(s)
- Describe what the dotplots tell you about the similarity between the sequences.
- Are there repeated regions, if so how many, and how long is the repeat?
- Do the conserved regions correspond to known functional regions (check annotation and CDD)?

2. Alignments

- Show the best local alignment, and best global alignment for each pair. Why is this the best alignment (what criterion did you use)?
- Describe how the dotplots and the two alignments are consistent/inconsistent with each other.
- If there are repeats, show the aligned repeats as a multiple alignment (this does not necessarily require using a multiple alignment program).
 - One program that could help is lalign
(https://fasta.bioch.virginia.edu/fasta_www2/fasta_www.cgi?rm=lalign&pgm=lal)
 - Another approach is to use a multiple alignment program to align repeats
 - Or you can do it by hand
- Do the conserved regions correspond to known functional regions
- Compare the alignments to the dotplots

BP2: Multiple Alignments/trees

50 pts. Protein vs DNA taxonomy

- 10 pts. For a group of organisms with known taxonomy, for instance orders of mammals, birds, or insects, choose at least 30 orthologous sequences (your choice); both DNA and protein sequences must be available. The easiest way to do this is probably using the Blast taxonomy report. Choose sequences for which both DNA and protein sequences are available, make sure the sequences are about the same length.
 - Relabel your sequences so they have informative names, e.g., H_sapiens instead of NC_0000712
- 10 pts. Make a multiple alignment for each set of sequences, using a program of your choice.
- 10 pts. Construct a Neighbor-joining tree for each alignment. A possible resource is <http://www.phylogeny.fr>
 - Remove unreliable parts and/or gaps from the alignment (Gblocks or other)
 - Use 1000 cycles of bootstrapping, or use LRT (faster)
- 20 pts. Compare the protein and DNA trees.
 - How do they differ? Comment on why they are different.
 - Are the differences correlated with the bootstrap values (do the differences in the trees occur at nodes with low bootstrap support)
 - Using a bootstrap cutoff of 70%, show the consensus trees for both sets of sequences.

BP3: Gene Finding with Blast

Up to 10 pts per gene.

For the DNA sequence file in 24bioproject3.fa (attached), use a combination of blastx, blastp, and blastp to identify all of the protein coding genes in the region. Use the model organisms (landmark) database for the protein analysis, and the refseq genome data base for nucleotides. For each gene, estimate

- start position
- end position
- number of exons, with start and end position for each
- possible function based on consensus of blast matches

BP4: Gene annotation (20pts/gene)

The files f12spades.gt1000.aug.fa and augustus.gff contain a newly assembled fungal genome and automatic annotation produced by Augustus. For each gene try to verify

- whether the gene model is correct, and if incorrect, the corrected gene model
- the function of the predicted gene
- gene ontology terms that should be associated with the gene. Include both the gene ontology number and the short description, for example “GO:1901576 organic substance biosynthetic process”

Include sufficient supporting evidence/arguments to allow your conclusions to be evaluated.

Software you might use includes Blast, Interpro, Egglog, and other gene prediction programs. You are not limited to these programs. Up to 20 points for each gene

BP5: PFam vs Blast (15 points/gene)

For a HMM of your choice, look up the matching sequences in a particular proteome (your choice) Report the chosen HMM, proteome, and number of matching proteins.

- For a selection of the FastA sequences, perform Blast searches and compare the results. The Blast search may contain duplicate sequences so this requires more than just counting. The number of duplicate sequences can be minimized by using the RefSeq protein database limited to your proteome of interest.
- Does the PFam list seem complete, or is the Blast search more complete?
- What Blast e-value threshold would give approximately the same list of sequences as in the PFam list?

Include sufficient supporting evidence/arguments to allow your conclusions to be evaluated.

BP6: Debruijn Graph assembly (75 points)

I have “sequenced” a famous quote. The sequence was produced in 30 character reads to a sampling depth of about 7X. At this coverage, you cannot expect to assemble a single contig, but you should be able to get a sufficient assembly to identify the quote. The data below is the list of 5mer words from the original reads. counts indicates the number of times the kmer was sampled. There are no errors in the data. This can easily be done by hand

- create the DBG, submit the graph after condensing the unbranched regions (i suggest powerpoint)
- submit the list of assembled contigs
- Determine the N50 of the assemble

kmer	count	kmer	count	kmer	count	kmer	count
ached	5	fires	8	notwi	6	thati	10
adeth	10	fromt	9	nthec	9	thatw	13
adows	5	frost	4	oesno	10	theas	7

afire	8	gdoes	8	okena	7	thecr	9
again	1	ghtfr	6	okent	11	thefr	4
aligh	6	gliitt	4	olddo	3	theol	11
allbe	16	goldd	3	oldth	10	therd	7
allsp	5	grene	6	omthe	10	thesh	4
allth	6	hadow	5	ongdo	8	those	6
ander	7	hallb	15	ootsa	5	tisgo	3
arelo	11	halls	5	osewh	7	tisst	10
areno	5	hatis	10	ostfr	4	treac	4
asbro	11	hatwa	13	ostth	11	trong	10
ashes	7	heash	7	otall	4	tsare	5
atisg	3	hecro	8	otgli	4	ttern	5
atiss	8	hedby	6	otrea	5	ttheo	11
atwas	13	hefro	4	otsar	5	twasb	13
bebla	8	heold	10	otwit	6	twith	5
bewok	8	herde	8	owand	8	wande	7
blade	8	hesaf	9	ownle	6	wasbr	11
broke	11	hesha	5	owssh	4	wedsh	6
bythe	5	hosew	7	pring	4	whowa	7
chedb	6	howan	8	proot	8	withe	5
crown	7	htfro	6	rarel	11	wnles	6
dbyth	6	ightf	6	rdeep	7	woken	7
ddoes	3	ingre	6	reach	4	wssha	4
deepr	7	iresh	8	relor	11	ythef	4
derar	9	isgol	3	renew	6		
detha	11	isstr	11	renot	5		
doesn	9	ither	6	resha	8		
dowss	4	itter	5	ringr	4		
dshal	5	kenal	8	rnota	5		
dthat	9	kenth	11	roken	11		
eache	5	ladet	10	romth	8		
eashe	7	lbebl	8	rond	10		
eblad	8	lbewo	9	roots	6		
ecrow	7	lddoe	3	rostf	4		
edbyt	7	ldtha	10	rownl	7		
edsha	7	lessa	6	safir	9		
eepro	7	light	6	sagai	2		
efros	4	litte	4	saren	5		
elost	11	llbeb	7	sbrok	11		
enali	8	llbew	9	sewho	7		
enewe	6	llspr	4	sgold	3		
enotr	5	lltho	5	shado	5		
enthe	10	lostt	11	shall	18		

eoldt	9	lspri	4	shesa	8
eproo	8	lthat	2	snotg	4
erare	9	lthos	5	snotw	6
erdee	7	mthea	7	sprin	4
ernot	5	mthes	4	ssaga	3
esafi	9	nalig	6	sshal	4
eshad	5	ndera	9	sstro	10
eshal	8	newed	6	stfro	5
esnot	10	ngdoe	8	stron	10
essag	4	ngren	6	stthe	12
ethat	12	nless	6	tallt	5
eweds	6	notal	6	terno	5
ewhow	7	notgl	4	tfrom	10
ewoke	8	notre	5	tglit	4

hints

- work in excel, building up the contigs as you work through the kmer list
- sort the kmers into an alphabetical list before starting to make it easier to find overlaps
- starting at an arbitrary kmer (or a tip if you like), start to build a condensed node, as you use each kmer, mark it as done (i use yellow shading)
- when you reach a branch point you will have to start new contigs (or technically condensed nodes), mark which nodes connect to each as you move on to new condensed nodes.